

Appendix 2.1-1 one obtains the following approximate values for the angles: $\Omega^0 = -0.3 \text{ gon} = -16'$, $\Phi^0 = -0.8 \text{ gon} = -43'$ and $K^0 = 298.7 \text{ gon} = 268^\circ 50'$.

The solution using the three-dimensional affine transformation has two disadvantages: on the one hand, the method breaks down when the four control points lie in a plane and, on the other hand, in the presence of only three control points, which is the basic requirement for an absolute orientation, it cannot be employed. A solution free of these disadvantages can be found by creating orthogonal vectors in both coordinate systems from three full control points. Details of this procedure are to be found in Section B 4.1.2.2 in Volume 2. Another method, with a detailed example, is included in Section B 5.1.1.3, again in Volume 2, which solves for the absolute orientation of near-vertical photographs without the need for initial approximate values.

4.5 Image coordinate refinement

Image coordinates taken from photographs must be corrected in different respects. This correction is known as image coordinate refinement. We have already learnt of a number of corrections:

- correction for lens distortion (Equation (3.1-3)).
- correction for film deformation in metric cameras (Section 3.2.1.2) and in semi-metric cameras (Section 3.8.4).
- correction for aplanarity (lack of flatness) of the CCD detector array (Section 3.3.4)
- correction for positional errors of detectors in CCD cameras or in film scanners (Section 3.4.3).

To meet demands for high accuracy one must also eliminate the influence of atmospheric refraction and take into account the curvature of the Earth.

4.5.1 Refraction correction for near-vertical photographs

Because the atmospheric pressure, the temperature and the humidity vary along the path of a ray of light, so too does the index of refraction, which is dependent on these three quantities. This means that the rays which give rise to a photograph are bent. The position of the image point P' must, therefore, be corrected by an amount $\Delta\rho$ in order to get the hypothetical metric image which would have been produced by central projection (Figure 4.5-1).

In image space, the tangent to the curved light ray will appear to represent the direction OP . In the case of a near-vertical photograph the angle $\Delta\tau$ between the light ray and

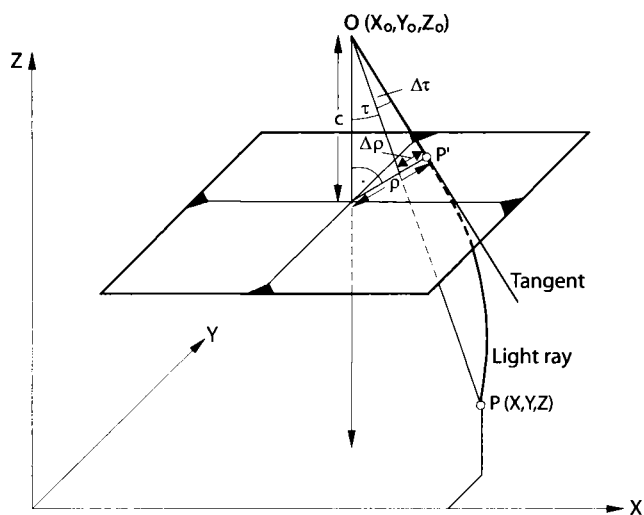


Figure 4.5-1: The effect of atmospheric refraction on a vertical photograph

the straight line OP can be represented with sufficient accuracy by a function of the angle τ and a coefficient K :

$$\Delta\tau = K \tan \tau = K \frac{\rho}{c} \quad (4.5-1)$$

It goes without saying that the coefficient K will vary according to the meteorologic data at the time of a particular photograph and, since the index of refraction varies with the wavelength, also with the wavelength to which the sensor is responsive. For a normal atmosphere stratified parallel to the surface of the Earth and for light in the visible wavelengths, K can be estimated as follows:¹⁵

$$K = 0.00241 \left(\frac{Z_0}{Z_0^2 - 6Z_0 + 250} - \frac{Z^2}{Z_0(Z^2 - 6Z + 250)} \right) \quad (4.5-2)$$

in which Z_0 = flying height above sea level (in [km])
and Z = height of the ground (in [km])

Normally it suffices to use approximate values of Z_0 and Z representative of the whole area being surveyed.

¹⁵Harris, W.D., Tewinkel, S.C. and Whitten, C.A.: Analytic Aerotriangulation. Coast and Geodetic Survey, Technical Bulletin No. 21, July 1962, corrected July 1963. Schut, G.H.: Ph.Eng. 35, pp. 79–86, 1969.

It remains only to give the relationship using which the radial correction $\Delta\rho$ can be found from $\Delta\tau$ (Figure 4.5-1):

$$\Delta\rho \cos \tau \approx \Delta\tau \sqrt{c^2 + \rho^2}$$

$$\Delta\rho \approx \frac{c^2 + \rho^2}{c} \Delta\tau$$

Taking Equation (4.5-1) into account:

$$\Delta\rho \approx \rho \left(1 + \frac{\rho^2}{c^2} \right) K \quad (4.5-3)$$

Finally, using the relationship (3.1-2), the correction $\Delta\rho$ can be resolved in the two coordinate directions ξ and η .

In Table 4.5-1 the expressions (4.5-2) and (4.5-3) are evaluated for particular cameras, photo scales and radial distances.

Photo scale	c [mm]	Z_0 [km]	Correction $\Delta\rho$ [μm]	
			$\rho = 90$ mm	$\rho = 130$ mm
1 : 10000	300	3.5	3	5
	150	2.0	2	4
	85	1.3	2	5
1 : 30000	300	9.5	8	12
	150	5.0	6	11
	85	3.0	6	13
1 : 100000	85	9.0	15	34
1 : 800000	300	240	1	2

Table 4.5-1: Correction of image coordinates for the effect of atmospheric refraction (Ground height $Z = 0.5$ km)

One sees from this table that the corrections reach a sufficient number of micrometers to be taken into account in processing only in small-scale photographs, above all in the case of super-wide photography. In the case of very precise aerotriangulation for larger-scale projects, however, refraction also needs to be considered. The last row shows that satellite images are not adversely affected by refraction.

In conclusion it must be mentioned that the Formulae (4.5-2) and (4.5-3) are valid only in the case of near-vertical photographs. With larger tilts it is necessary to move to more general formulae¹⁶.

Exercise 4.5-1. An aerial photograph ($c = 85$ mm) was taken from a height of 9100 m above sea level. Calculate corrections for the effect of atmospheric refraction at 25 points of a $5 \text{ cm} \times 5 \text{ cm}$ rectangular raster. In the left-hand side of the image the ground

¹⁶See, for example, Rinner, K.: JEK, Band IIIa/1, §§ 22–23.

is flat and level with a height of 600 m. It rises from the mid-line of the photographs with a slope of 30%. (Answer: In two corners of the picture $\Delta\rho$ reaches $37\text{ }\mu\text{m}$; in the other two corners $\Delta\rho = 42\text{ }\mu\text{m}$.)

An important special case concerns horizontal photographs in terrestrial photogrammetry. In this case the influence of atmospheric refraction is usually dealt with together with Earth curvature, although treating it separately is didactically more correct. Refraction is actually a physical error in the formation of the image while Earth curvature is a geometrical “error” in the ground coordinates.

4.5.2 Correction for refraction and Earth curvature in horizontal photographs

As has already been indicated, the effect of Earth curvature is not a photogrammetric error but is related simply to the definition of coordinate systems. Photogrammetry provides coordinates in a three-dimensional Cartesian coordinate system in which the position of an object point is fixed by orthogonal projection on the respective coordinate planes. In the field survey system, on the other hand, heights of ground points are referred to the ellipsoid while the XY coordinates are more or less distorted by the mathematical projection of the ellipsoid onto a plane. For this reason the ground survey coordinates of control points should first be transformed into three-dimensional Cartesian coordinates before the start of the photogrammetric work.

Fortunately, however, the differences between the ground survey coordinate system and a three-dimensional Cartesian coordinate system are small so that in practice the problem can be simplified. In Volume 2, Section B 5.4, however, a full discussion appears. Here and in the following section the discussion is restricted to approximate solutions for the coordinate most strongly affected, namely the ground height.

First, the effect of Earth curvature combined with atmospheric refraction for horizontal photographs is dealt with. The Earth is regarded as a sphere of radius $R = 6370\text{ km}$. The symbol $\bar{Z}$ is used for heights referred to the sphere as opposed to Z for heights referred to the three-dimensional Cartesian system (Figure 4.5-2).

The effects of Earth curvature ΔE for the point P are given by the well-known formula:

$$\Delta E \approx \frac{A^2}{2R} \quad (4.5-4)$$

Standard textbooks on surveying¹⁷ give an approximate equation¹⁸ for trigonometric height measurement which takes account of Earth curvature and the influence of refraction:

¹⁷For example Kahmen, H., Faig, W.: Surveying. 578 p., Walter de Gruyter, Berlin, 1988.

¹⁸The approximation in this relation is too crude for large distances and large height differences, conditions which, however, seldom arise in photogrammetric practice.

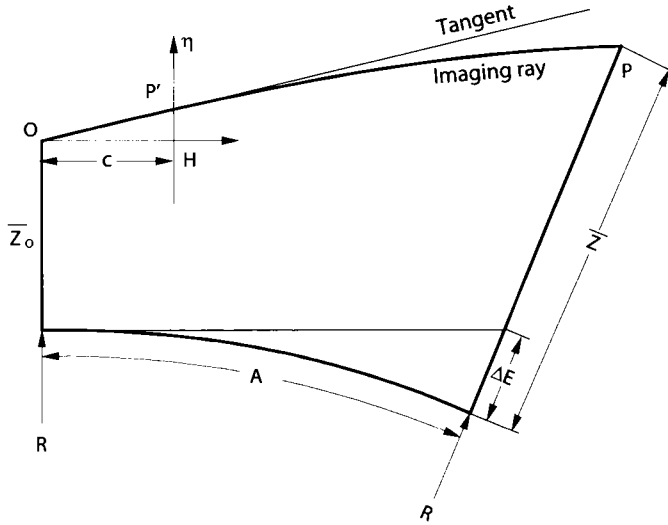


Figure 4.5-2: Atmospheric refraction and Earth curvature for a horizontal photograph

$$\bar{Z} = \bar{Z}_0 + A \tan \alpha + A \tan \alpha \frac{\bar{Z}_0 + \bar{Z}}{2R} + (1 - k) \frac{A^2}{2R} \quad (4.5-5)$$

in which k ... refraction coefficient, usually taken as 0.13

For photogrammetric point determination with horizontal photographs $\tan \alpha$ may be replaced by the ratio η/c :

$$\bar{Z} = \bar{Z}_0 + \eta \frac{A}{c} + \eta \frac{A}{c} \frac{\bar{Z}_0 + \bar{Z}}{2R} + (1 - k) \frac{A^2}{2R} \quad (4.5-6)$$

What is required, however, for use with the three-dimensional object coordinate system is the significantly simpler relation:

$$\bar{Z} = \bar{Z}_0 + \frac{A}{c} (\eta + \Delta\eta) \quad (4.5-7)$$

To compensate for the use of heights referred to the sphere rather than Z coordinates from the Cartesian coordinate system, the correction $\Delta\eta$ is introduced to the image coordinates η . Subtraction of Equation (4.5-7) from (4.5-6) gives:

$$\frac{A}{c} \Delta\eta = \eta \frac{A}{c} \frac{\bar{Z}_0 + \bar{Z}}{2R} + (1 - k) \frac{A^2}{2R}$$

Finally, re-arranging gives the desired formula:

$$\Delta\eta = \eta \frac{\bar{Z}_0 + \bar{Z}}{2R} + (1 - k) \frac{Ac}{2R} \quad (4.5-8)$$

If the image coordinates η are corrected by an amount $\Delta\eta$ and if the photogrammetric measurements are made in the usual way, then the height coordinates, whether of control points or of new points, will be referred to the sphere; they will also be free of the influence of refraction. For the computation of the correction $\Delta\eta$, however, the distance A and the value $\bar{Z}$ are necessary. If P is a new point the following procedure, suitable for automation, is recommended:

- first measure the point photogrammetrically without correction for Earth curvature and refraction
- using the result so obtained, compute A , $\bar{Z}$ (with adequate accuracy) and $\Delta\eta$
- finally, repeat the photogrammetric computation using the corrected image coordinates.

Table 4.5-2 gives some typical values for the correction $\Delta\eta$.

A [km]	0.1	0.25	0.5	1	5	20
$\Delta\eta$ [μm]	0.7	1.7	3.4	6.8	34	137

Table 4.5-2: Corrections, $\Delta\eta$, to image coordinates for the influence of refraction and Earth curvature ($c = 100$ mm, $k = 0.13$, $R = 6370$ km, $\eta = 0$)

It will be seen that, up to a few hundred metres object distance the correction is smaller than the measuring accuracy of a few micrometres, but may not be ignored for distances greater than about 1 km.

Exercise 4.5-2. Compute the correction $\Delta\eta$ and the $\bar{Z}$ coordinate for an extreme case with $A = 5000$ m, $\bar{Z}_0 = 1000$ m, $\eta = 30$ mm, $c = 100$ mm. (Result: $\Delta\eta = 42.4$ μm , $\bar{Z} = 2502.12$ m). Additional exercise: Compute the $\bar{Z}$ coordinate using a refraction coefficient of 0.10. (Result: $\Delta\eta = 43.6$ μm , $\bar{Z} = 2502.18$.)

Exercise 4.5-3. Using the data of Table 4.5-2 compute the standard errors of both the correction $\Delta\eta$ and the ground height Z under the assumption that the refraction coefficient $k = 0.13$ has a standard error of ± 0.03 . (Solution: for example, if $A = 5$ km: $\sigma_{\Delta\eta} = \pm 1.2$ μm , $\sigma_{\bar{Z}} = \pm 5.9$ cm.)

4.5.3 Earth curvature correction for near-vertical photographs

In this section the influence of and correction for Earth curvature in near-vertical photographs is considered. A stereomodel produced photogrammetrically in the usual manner from two aerial photographs is referred to the three-dimensional Cartesian coordinate system XYZ (Figure 4.5-3). Heights from land survey, however, refer to a sphere with the Earth radius R .

Earth curvature can be taken into account in the photogrammetric process as now described. Control point heights $\bar{Z}$, referred to the curved datum surface $\bar{Z}_0 = 0$, are

So that an assessment may be made of the extent to which a correction for Earth curvature is important, the maximum correction $\Delta E_{\max}$ in parts per thousand of the flying height is given in Table 4.5-3 for various photo scales. The value $\Delta E_{\max}$ (Figure 4.5-3) comes from Equation (4.5-4) in which $B\sqrt{5}/2$, the semi-diagonal of the model, is substituted for A :

$$\Delta E_{\max} = \frac{\left(\frac{B}{2}\sqrt{5}\right)^2}{2 R} = \frac{5 B^2}{8 R} \quad (4.5-9)$$

In view of photogrammetric height accuracy of 0.06‰ for a measured point (Section 4.6), Table 4.5-3 shows that Earth curvature must be taken into account:

- for super-wide angle photographs from a photo scale of about 1 : 6000,
- for wide angle photographs from a photo scale of about 1 : 10000 and
- for normal angle photographs from a photo scale of about 1 : 20000.

Photo scale	Maximum Earth curvature correction $\Delta E_{\max}$			
	[m]	(‰ of the flying height)		
		$c = 85 \text{ mm}$	$c = 150 \text{ mm}$	$c = 300 \text{ mm}$
1:10000	0.08	0.10	0.06	0.03
1:50000	2.1	0.49	0.28	0.14
1:100000	8.3	0.98	0.56	0.28

Table 4.5-3: Maximum Earth curvature correction $\Delta E_{\max}$ at the corners of a stereo-model with 60% forward overlap

4.5.4 Virtual (digital) correction image

In digital photogrammetry every completed correction to be implemented in the context of image coordinate refinement entails a resampling, that is a restructuring of the pixels in a new orthogonal raster (Section 2.2.3). As a rule every resampling is accompanied by loss of information. In view of the relatively large number of corrections which mount up in the course of image coordinate refinement a solution is sought which, if possible, manages with one single resampling.

With a virtual correction image¹⁹ one achieves this aim. The procedure is explained with reference to Figure 4.5-4; it is a typical example which can readily be generalized. Let the digitized photograph have an arbitrary $\xi'\eta'$ coordinate system (Section 3.4). A corrected image matrix is required in the $\xi\eta$ system, the coordinate system of a digital metric camera or a digital metric image (Figures 4.5-4 and 3.1-10). As in indirect resampling (Figure 2.2-5), an orthogonal raster is first defined in the digital image

¹⁹The idea was adopted from T. Schenk (Digital Photogrammetry. TerraScience, Volume I, 1999).